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United States Patent Application Publication	20250278141
Kind Code	A1
Publication Date	September 04, 2025
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PEN

Abstract

A pen includes a vibration device including a vibrator, and a processing unit that suppresses movement of the vibrator during a pen non-use period during which no pen input is being performed.

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Family ID: 91588786

Appl. No.: 19/209524

Filed: May 15, 2025

Foreign Application Priority Data

JP	2022-203167	Dec. 20, 2022
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Related U.S. Application Data

parent WO continuation PCT/JP2023/044843 20231214 PENDING child US 19209524

Publication Classification

Int. Cl.: G06F3/01 (20060101); G06F3/0354 (20130101); G06F3/038 (20130101)

U.S. Cl.:

CPC G06F3/016 (20130101); G06F3/03545 (20130101); G06F3/0383 (20130101);

Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to a pen, and particularly to a pen having a haptic function.

Description of the Related Art

[0002] Some pens used to perform pen input on a panel surface have a function that reproduces a tactile sensation felt when using a traditional pen such as a ballpoint pen or a pencil (hereinafter referred to as a “haptic function”). This type of pen typically has a built-in vibration device, and is configured to vibrate the vibration device depending on whether the pen tip is in contact with the panel surface, whether the pen tip is traveling on the panel surface, etc. Japanese Patent Laid-open No. 2014-222492 discloses an example of a pen having such configuration.

[0003] When the tip of a pen having a built-in vibration device is brought into contact with a panel surface, there are cases where the vibration device may vibrate due to the impact of the contact, even though there is no electrical control to make the device vibrate. This is because the vibrator inside the vibration device moves due to the physical impact. Such vibration gives an uncomfortable feeling to the user and should be suppressed.

BRIEF SUMMARY

[0004] According to one aspect, a pen is provided that can suppress vibration of a vibration device caused by an impact when the pen tip is brought into contact with a panel surface.

[0005] The present disclosure relates to a pen including a vibration device containing a vibrator, and a processing unit that suppresses movement of the vibrator during a pen non-use period during which no pen input is being performed.

[0006] According to the present disclosure, the movement of the vibrator is suppressed during the pen non-use period, so as to suppress vibration of the vibration device caused by the impact generated when the pen tip is brought into contact with the panel surface.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0007] FIG. 1 is a diagram illustrating the system configuration of a position detecting system according to a first embodiment of the present disclosure;

[0008] FIGS. 2A and 2B are diagrams illustrating the structure of a vibration device;

[0009] FIG. 3 is a diagram illustrating functional blocks of a pen and an electronic device;

[0010] FIG. 4 is a process flow diagram illustrating the control of the vibration device executed by a short-range wireless communication unit according to the first embodiment of the present disclosure;

[0011] FIG. 5 is a diagram illustrating the functional blocks of the pen and the electronic device according to a second embodiment of the present disclosure;

[0012] FIG. 6 is a process flow diagram illustrating the control of the vibration device executed by a short-range wireless communication unit according to the second embodiment of the present disclosure;

[0013] FIG. 7 is a diagram illustrating functional blocks of the pen according to a third embodiment of the present disclosure;

[0014] FIG. 8 is a process flow diagram illustrating the control of the vibration device executed by a short-range wireless communication unit according to the third embodiment of the present disclosure;

[0015] FIG. 9 is a process flow diagram illustrating the control of the vibration device executed by a short-range wireless communication unit according to a fourth embodiment of the present disclosure; and

[0016] FIG. 10 is a process flow diagram illustrating the control of the vibration device executed by a short-range wireless communication unit according to the background art of the present disclosure.

DETAILED DESCRIPTION

[0017] Embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

[0018] FIG. 1 is a diagram illustrating the system configuration of a position detecting system 1 according to a first embodiment of the present disclosure. As illustrated in the figure, the position detecting system 1 includes a pen 2 and an electronic device 3.

[0019] The pen 2 is an active pen compatible with an active capacitance method, and includes a pen tip electrode 20, a vibration device 21, and a side switch 22. In the present embodiment, the description will continue assuming that the pen 2 is an active pen, but other types of pens can also be used as the pen 2, such as an electromagnetic induction pen compatible with an electromagnetic induction method (EMR method).

[0020] The pen tip electrode 20 is a conductor disposed near the pen tip of the pen 2. The pen 2, which is an active pen, is configured to use this pen tip electrode 20 to perform bidirectional communication with a sensor controller 32 provided in the electronic device 3. Hereinafter, a signal transmitted from the sensor controller 32 to the pen 2 through this communication will be referred to as an “uplink signal US,” and a signal transmitted from the pen 2 to the sensor controller 32 through this communication will be referred to as a “downlink signal DS.”

[0021] The uplink signal US is a signal modulated by data to be transmitted from the sensor controller 32 to the pen 2. The data includes data indicating the traveling speed of the pen 2 on the panel surface as well as commands that define the operation of the pen 2. The downlink signal DS includes an unmodulated burst signal and a data signal modulated by data to be transmitted from the pen 2 to the sensor controller 32. The burst signal is used by the sensor controller 32 to detect the position of the pen 2. The data transmitted as the data signal includes data concerning a writing pressure value indicating the pressure applied to the pen tip, on/off information of the side switch 22, a pen identification (ID) previously assigned to the pen 2, etc.

[0022] The vibration device 21 is a device including a vibrator that vibrates in response to an electric signal, and is typically a linear resonant actuator (LRA). However, the present disclosure is also applicable to the pen 2 having a vibration device other than a linear resonant actuator (LRA).

[0023] FIGS. 2A and 2B are diagrams illustrating an example of the structure of the vibration device 21. FIG. 2A is a cross-sectional view of the vibration device 21 taken along the XY plane, and FIG. 2B is a cross-sectional view of the vibration device 21 taken along the XZ plane. As illustrated in FIGS. 2A and 2B, the vibration device 21, which is a linear resonant actuator, has a configuration in which a rectangular parallelepiped-shaped vibrator 21b, leaf springs 21c and 21d, and a coil 21e are arranged in a rectangular parallelepiped-shaped housing 21a.

[0024] The vibrator 21b is a permanent magnet having an S pole at one end in an X direction and an N pole at the other end, and is so arranged as to be able to move in the illustrated A direction (parallel to the X direction). The leaf spring 21c is arranged between a surface on one side of the vibrator 21b in the X direction and the inner surface of the housing 21a facing the surface on one side. Further, the leaf spring 21d is arranged between the surface on the other side of the vibrator 21b in the X direction and the inner surface of the housing 21a facing the surface on the other side. With this arrangement, the leaf springs 21c and 21d each play a role of biasing the vibrator 21b toward the inside of the housing 21a.

[0025] The coil 21e is a member that generates an alternating magnetic field in response to an electric signal (alternating current) supplied from the outside, and is disposed below the vibrator 21b in a Z direction. In the coil 21e, the magnetic field lines generated at the center of the coil 21e when a current is passed through the coil 21e would have a direction in parallel to the Z direction. When an alternating magnetic field is generated in the coil 21e, force corresponding to the direction

of the magnetic field being generated is applied to the vibrator **21b**. Due to the force thus applied and the biasing force of the leaf springs **21c** and **21d**, the vibrator **21b** vibrates in the illustrated direction A when an electric signal is supplied to the coil **21e** from the outside. The haptic function of the pen **2** is achieved by the vibration of the vibrator **21b**.

[0026] Referring back to FIG. 1, the electronic device **3** is a computer that supports pen input to a panel surface **3a**, and includes a display device **30**, a sensor **31**, the sensor controller **32**, a short-range wireless communication unit **33**, and a host processor **34**. In a typical example, the electronic device **3** is a smartphone or a tablet terminal.

[0027] The display device **30** is a device that displays an image on the panel surface **3a** in accordance with a video signal supplied from the host processor **34**, and is configured of, for example, a liquid crystal display or an organic electroluminescence (EL) display.

[0028] The sensor **31** is a device having a plurality of linear electrodes each extending inside the panel surface **3a**. Each linear electrode is made of a transparent conductor and is disposed on the display surface of the display device **30**. This allows pen input to be carried out on the display surface of the electronic device **3**. However, the electronic device **3** may be an electronic device (such as a digitizer) that performs pen input on a surface that is not a display surface.

[0029] The sensor controller **32** is an integrated circuit having functions of detecting a position of the pen **2** on the panel surface **3a**, receiving data transmitted by the pen **2**, and supplying the detected position data and received data to the host processor **34**. Specifically, the sensor controller **32** periodically transmits uplink signal US by generating the uplink signal US including a command for the pen **2** and supplying the uplink signal US to some or all of the plurality of linear electrodes. The sensor controller **32** receives downlink signal DS transmitted from the pen **2** in response to the uplink signal US each time the uplink signal US is transmitted. Specifically, the sensor controller **32** receives the burst signal constituting the downlink signal DS at each of the plurality of linear electrodes, and detects the position of the pen **2** according to the reception intensity at each linear electrode. The sensor controller **32** acquires the data transmitted from the pen **2** by demodulating the downlink signal DS received at any one of the linear electrodes.

[0030] The sensor controller **32** also performs a process of deriving the traveling speed of the pen **2** on the panel surface **3a** in reference to the history of the detected positions of the pen **2**. The sensor controller **32** is configured to include data indicating the derived traveling speed in the uplink signal US together with the command.

[0031] The short-range wireless communication unit **33** is, for example, a communication device that performs bidirectional communication with other devices by short-range wireless communication BT such as the Bluetooth (registered trademark) communication standard. As will be described in detail later, the pen **2** is compatible with the short-range wireless communication BT also, and the short-range wireless communication unit **33** performs communication with the pen **2** by the short-range wireless communication BT under the control of the host processor **34**.

[0032] The host processor **34** is a central processing unit of the electronic device **3** that plays a role of executing a program read from an unillustrated memory and executing the operating system and various applications of the electronic device **3**. The host processor **34** also has a role of supplying a video signal obtained as a result of the execution of the program to the display device **30**.

[0033] The applications executed by the host processor **34** include a drawing application that performs drawing according to the traveling trajectory of the pen **2**. The drawing application plays a role of executing a process of generating stroke data (data indicating a trajectory of positions) by using the position information and data supplied from the sensor controller **32**, a process of rendering the generated stroke data and displaying the data on the display device **30**, a process of generating and recording digital ink including the generated stroke data, and a process of transmitting the generated digital ink to an external device.

[0034] The drawing application is configured to be able to set information required for rendering the generated stroke data, through user operation (hereinafter referred to as “rendering

information”). Specific examples of the rendering information include information regarding the type and size of the brush (pen tip), the width of the line that is input, etc. The rendering information that has been set is used by the drawing application for rendering the stroke data, and is also used to vibrate the vibration device **21** in the pen **2**.

[0035] The latter point will be described in detail. The drawing application stores in advance a table that associates rendering information with identification information. The identification information identifies the waveform pattern of an electric signal (alternating current) supplied to the vibration device **21**. When the drawing application is started or when the rendering information is changed, the drawing application performs a process of reading from this table the identification information corresponding to the currently set rendering information. The drawing application generates a control signal indicative of the read-out identification information, and uses the short-range wireless communication unit **33** to transmit the control signal via the short-range wireless communication BT to the pen **2**. While details of the process executed by the pen **2** that has received the control signal transmitted in this manner will be described later, this results in, while the user is writing with the pen **2**, the vibration device **21** vibrating according to the waveform pattern indicated by the transmitted identification information.

[0036] FIG. **3** is a diagram illustrating functional blocks of the pen **2** and the electronic device **3**. The functional blocks of the electronic device **3** illustrated in this figure are the same as those illustrated in FIG. **1**. However, the display device **30** is omitted.

[0037] As illustrated in FIG. **3**, the pen **2** includes a core body **23**, a ring electrode **24**, a pressure sensor **25**, a pen processing unit **26**, a capacitive sensor **27**, a short-range wireless communication unit **28**, and a haptics driver **29** as well as the pen tip electrode **20**, the vibration device **21**, and the side switch **22** that are also illustrated in FIG. **1**.

[0038] The core body **23** is a substantially cylindrical member that constitutes the pen tip of the pen **2**. The pen tip electrode **20** is formed at the tip of the core body **23**. The ring electrode **24** is a ring-shaped conductor disposed in the middle of the core body **23**. The function of the ring electrode **24** will be described later.

[0039] The tip of the core body **23** protrudes from the tip of the pen **2**, and the rear end of the core body **23** is in contact with the pressure sensor **25**. The pressure sensor **25** is a sensor that detects the pressure applied to the tip (pen tip) of the core body **23** through this contact, and is configured to supply a writing pressure value indicating the detected pressure to the pen processing unit **26**.

[0040] The side switch **22** is an assembly of one or more switches each of which can be turned on and off by the user. On/off information regarding each switch is supplied to the pen processing unit **26**.

[0041] The pen processing unit **26** is an integrated circuit that performs processing of receiving the uplink signal US via the pen tip electrode **20** or the ring electrode **24**, and also generating the downlink signal DS according to the received uplink signal US and transmitting the signal via the pen tip electrode **20**. The pen processing unit **26** is configured to generate the downlink signal DS to be transmitted, in accordance with a command indicated by the received uplink signal US. The downlink signal DS thus generated includes an unmodulated burst signal, and data signals including a writing pressure value supplied from the pressure sensor **25** to the pen processing unit **26** and on/off information regarding each switch supplied from the side switch **22** to the pen processing unit **26**.

[0042] There are also cases where the pen processing unit **26** may transmit the downlink signal DS from the ring electrode **24**. This downlink signal DS may be a simple burst signal, or may be a signal including a burst signal and a data signal, similarly to the downlink signal DS transmitted from the pen tip electrode **20**. The burst signal transmitted from the ring electrode **24** is used by the sensor controller **32** to detect the inclination of the pen **2**.

[0043] The pen processing unit **26** performs processing to supply data indicating the writing pressure value supplied from the pressure sensor **25** and the traveling speed of the pen **2** contained

in the received uplink signal US to the short-range wireless communication unit **28**. The processing performed by the short-range wireless communication unit **28** upon receiving this supply will be described later.

[0044] The capacitive sensor **27** is a capacitance type sensor disposed on the surface (side surface) of the pen **2**, and serves to detect the user's finger. The capacitive sensor **27** performs processing to detect the user's finger according to control of the short-range wireless communication unit **28**. The detection result is supplied from the capacitive sensor **27** to the short-range wireless communication unit **28**.

[0045] The short-range wireless communication unit **28** is a communication device that performs bidirectional communication with other devices via the short-range wireless communication BT, and is also a control device for the vibration device **21**. The short-range wireless communication unit **28** receives the above-mentioned control signal from the electronic device **3** via the short-range wireless communication BT, and plays a role of controlling the haptics driver **29** in response to the received control signal. The haptics driver **29** is a circuit that generates an electrical signal (alternating current) in response to the control signal and supplies the generated electrical signal to the coil **21e** of the vibration device **21**.

[0046] The control of the haptics driver **29** by the short-range wireless communication unit **28** will be described in more detail. The short-range wireless communication unit **28** stores in advance a table that associates identification information which identifies the waveform pattern of an electric signal (alternating current) supplied to the vibration device **21** with the waveform pattern itself. The short-range wireless communication unit **28** performs a process of reading from the above-mentioned table a waveform pattern that corresponds to the identification information indicated by the control signal received from the electronic device **3**, to thereby determine the waveform pattern to be supplied to the haptics driver **29**.

[0047] The short-range wireless communication unit **28** performs a process of determining whether or not to vibrate the vibration device **21** in reference to the data indicating the writing pressure value and the traveling speed of the pen **2** supplied from the pen processing unit **26**. As a result of this determination, the short-range wireless communication unit **28** that has determined to vibrate the vibration device **21** performs a process of supplying a waveform pattern determined based on a control signal received from the electronic device **3** to the haptics driver **29**.

[0048] The haptics driver **29** generates an electric signal (first electric signal) that oscillates according to the waveform pattern supplied from the short-range wireless communication unit **28**, and supplies the electric signal to the coil **21e** of the vibration device **21**. This causes the vibration device **21** to vibrate with the waveform pattern designated by the electronic device **3**.

[0049] FIG. **10** is a process flow diagram illustrating the control of the vibration device **21** executed by the short-range wireless communication unit **28** according to the background art of the present disclosure. With reference to FIG. **10**, a problem of the control of the vibration device **21** executed by the short-range wireless communication unit **28** to which the present disclosure is not applied will be described in more detail.

[0050] Steps **S100** to **S102** indicated by broken lines are operations or processes performed outside the short-range wireless communication unit **28**. First, the user causes the pen tip of the pen **2** to touch the panel surface **3a** (step **S100**). As a result, the vibrator **21b** in the vibration device **21** moves due to the impact of the touch (step **S101**). One aspect of the present disclosure is to suppress the vibration of the vibration device **21** caused by this movement. When the pen tip comes into contact with the panel surface **3a**, the writing pressure value supplied from the pressure sensor **25** to the pen processing unit **26** becomes a value indicating a contact (e.g., a value greater than 0) (step **S102**). When the writing pressure value is a value indicating a contact, it is considered that “a writing pressure is on,” and when the writing pressure value is a value not indicating a contact (e.g., 0), it is considered that “a writing pressure is off.”

[0051] Steps **S103** and **S104** are processes for determining whether or not the user is writing with

the pen 2. Specifically, the short-range wireless communication unit 28 first determines whether or not the writing pressure is on by referring to the writing pressure value sequentially supplied from the pen processing unit 26 (step S103). If it is determined that the writing pressure is not on, the short-range wireless communication unit 28 determines that the user is not writing with the pen 2, and returns to step S100 to continue the processing. On the other hand, the short-range wireless communication unit 28 that has determined that the writing pressure is on determines whether or not the pen 2 is traveling by referring to data indicating the traveling speed of the pen 2 sequentially supplied from the pen processing unit 26 (step S104). If it is determined that the pen 2 is not moving (traveling), the short-range wireless communication unit 28 determines that the user is not writing with the pen 2, and returns to step S103 to continue the processing.

[0052] On the other hand, in the case of determining in step S104 that the pen 2 is traveling, the short-range wireless communication unit 28 determines that the user is writing with the pen 2. Then, the short-range wireless communication unit 28 determines a waveform pattern in reference to the control signal received from the electronic device 3 (step S105), and supplies the determined waveform pattern to the haptics driver 29 (step S106). As a result, since an electric signal that oscillates according to the determined waveform pattern is supplied from the haptics driver 29 to the vibration device 21, the vibration device 21 of the pen 2 is vibrated with a pattern according to the rendering information currently set in the drawing application while the user is writing with the pen 2. Such control of the vibration device 21 cannot suppress the movement of the vibrator 21b in step S101.

[0053] Now referring back to FIG. 3, in order to suppress the movement of the vibrator 21b in step S101 of FIG. 10, in addition to controlling the vibration device 21 illustrated in FIG. 10, the short-range wireless communication unit 28 according to the present disclosure performs control to suppress the movement of the vibrator 21b during the pen non-use period during which no pen input is performed. Specifically, the short-range wireless communication unit 28 supplies an electric signal for vibrating the vibrator 21b with an amplitude so small that the user cannot perceive the vibration (a second electric signal that oscillates with an amplitude smaller than that of the first electric signal described above) to the vibration device 21 during the pen non-use period, to thereby suppress the movement of the vibrator 21b during the pen non-use period. This prevents the vibrator 21b from moving freely, making it possible to suppress the vibration of the vibration device 21 caused by the impact generated when the pen tip is brought into contact with the panel surface 3a. FIG. 4 is a process flow diagram illustrating the control of the vibration device 21 executed by the short-range wireless communication unit 28 according to the present embodiment. As illustrated in FIG. 4, the short-range wireless communication unit 28 first determines whether or not the current time is within a pen non-use period (step S1). Specifically, the short-range wireless communication unit 28 may make the determination of step S1 by considering a period during which the writing pressure is off as the pen non-use period, or by considering a period during which the writing pressure is off and the pen 2 is not traveling as the pen non-use period.

Alternatively, the short-range wireless communication unit 28 may enter the pen non-use period based on an explicit operation performed by the user using the side switch 22, and make the determination of step S1 depending on whether or not the pen non-use period has been entered.

[0054] The short-range wireless communication unit 28 that has determined in step S1 that the current time is not within the pen non-use period executes the processes of steps S103 to S106 described with reference to FIG. 10. As a result, while the user is writing with the pen 2, the vibration device 21 of the pen 2 vibrates in patterns according to the rendering information currently set in the drawing application, as described above.

[0055] The short-range wireless communication unit 28 according to the present embodiment is configured to further determine, when a negative determination result is obtained in step S103 or step S104, whether there is any other basis (factor) for vibration (step S5). If there is another vibration factor, the process moves to step S105. This process enables generation of vibrations

other than those intended to reproduce the tactile sensation during a writing operation. Those other vibrations include, for example, vibrations as a warning alert regarding a remaining battery level or vibrations in response to an instruction received at an arbitrary timing from the electronic device **3**. The instruction received from the electronic device **3** may be transmitted via the short-range wireless communication BT or may be transmitted in the uplink signal US. The instruction may include identification information which identifies the waveform pattern of the electrical signal to be supplied to the vibration device **21**, and in this case, the waveform pattern determined in step **S105** may preferably be the waveform pattern indicated by the identification information.

[0056] When a negative determination result is obtained in step **S5**, or when the process of step **S106** (supplying the determined waveform pattern to the haptics driver **29**) is completed, the short-range wireless communication unit **28** returns to step **S1** to continue the processing. Note that, when some or all of the processes of steps **S103** and **S104** overlap with the process of step **S1**, the execution of the overlapping parts of steps **S103** and **S104** may be omitted.

[0057] On the other hand, the short-range wireless communication unit **28** that has determined in step **S1** that the current time is within the pen non-use period performs finger detection by using the capacitive sensor **27** (step **S2**), and determines whether a finger has been detected as a result (step **S3**). In the case where it is determined that a finger has not been detected, the short-range wireless communication unit **28** returns to step **S1** and continues the processing. On the other hand, when it is determined that a finger has been detected, the short-range wireless communication unit **28** supplies a waveform pattern for micro-vibration to the haptics driver **29** (step **S4**). This waveform pattern is also one of a plurality of waveform patterns stored in advance in the short-range wireless communication unit **28**, but unlike the waveform pattern for fulfilling the haptic function, this is a waveform pattern that oscillates with an amplitude so small that the user cannot sense or recognize the vibration. As a result, an electric signal for vibrating the vibrator **21b** with an amplitude so small that the user cannot sense the vibration is supplied from the haptics driver **29** to the vibration device **21** (to both ends of the coil **21e**), and consequently, the vibrator **21b** cannot move freely, so that the vibration of the vibration device **21** caused by the impact generated when the pen tip is brought into contact with the panel surface **3a** is suppressed. The short-range wireless communication unit **28** that has completed the supply of the waveform pattern in step **S4** returns to step **S1** to continue the processing.

[0058] As described above, according to the position detecting system **1** of the present embodiment, when the current time is within the pen non-use period and a finger is being detected by the capacitive sensor **27**, an electric signal corresponding to a waveform pattern for micro-vibration is supplied to the vibration device **21**, so that the movement of the vibrator **21b** during the pen non-use period is suppressed. Hence, the vibration of the vibration device **21** caused by the impact generated when the pen tip is brought into contact with the panel surface **3a** can be suppressed.

[0059] It is to be noted that, in the position detecting system **1** according to the present embodiment, the capacitive sensor **27** does not have to be provided in the pen **2**, and steps **S2** and **S3** in FIG. **4** may be omitted, and the position detecting system **1** still achieves the same effects as described above. However, in this case, the vibration device **21** needs to continue vibrating even when the user leaves the pen **2** on a desk, etc., and thus the power consumption of the pen **2** increases. It is therefore preferable to provide the pen **2** with the capacitive sensor **27** and execute steps **S2** and **S3** in FIG. **4** as in the present embodiment.

[0060] Next, the position detecting system **1** according to a second embodiment of the present disclosure will be described. The position detecting system **1** according to the present embodiment differs from the position detecting system **1** according to the first embodiment in that the pen **2** does not have the capacitive sensor **27** and the short-range wireless communication unit **28** has a sleep mode. The content of the control of the vibration device **21** executed by the short-range wireless communication unit **28** is also different. Otherwise, the position detecting system **1**

according to the second embodiment is similar to the position detecting system **1** according to the first embodiment. Below, the position detecting system **1** according to the present embodiment will be described, by focusing on the differences from the position detecting system **1** according to the first embodiment. FIG. **5** is a diagram illustrating functional blocks of the pen **2** and the electronic device **3** according to the present embodiment. As can be understood by comparing FIG. **5** with FIG. **3**, the pen **2** according to the present embodiment is not provided with the capacitive sensor **27**.

[0061] The short-range wireless communication unit **28** according to the present embodiment is configured to operate in either a startup mode or a sleep mode. The sleep mode is a mode in which no processing is performed other than processing for accepting a predetermined startup operation. The predetermined startup operation may be, for example, an operation of changing the writing pressure value to a value greater than 0 (i.e., an operation of causing the pen tip to come in contact with the panel surface **3a**), or an operation of pressing the side switch **22**. The startup mode is a mode entered when a predetermined startup operation is accepted, and the short-range wireless communication unit **28** that has entered the startup mode is in a state of being able to execute all functions including communicating with other devices via the short-range wireless communication BT, controlling the vibration device **21**, and controlling to suppress the movement of the vibrator **21b** during the pen non-use period.

[0062] FIG. **6** is a process flow diagram illustrating the control of the vibration device **21** executed by the short-range wireless communication unit **28** according to the present embodiment. As illustrated in FIG. **6**, the short-range wireless communication unit **28** first enters a sleep mode (step **S10**), and repeatedly executes a determination as to whether or not a predetermined startup operation has been executed, until a positive determination result is obtained (step **S11**).

[0063] In the case where a positive determination result is obtained in step **S11**, the short-range wireless communication unit **28** enters a startup mode (step **S12**) and determines whether or not the current time is within the pen non-use period (step **S13**). The details of this determination are similar to those in step **S1** of FIG. **4**.

[0064] The short-range wireless communication unit **28** that has determined in step **S13** that the current time is not within the pen non-use period executes the processes in steps **S103** to **S106** and step **S5**, as in the example of FIG. **4**. The details of the processing are similar to those in the case where it is determined in step **S1** of FIG. **4** that the current time is not within the pen non-use period.

[0065] On the other hand, the short-range wireless communication unit **28** that has determined in step **S13** that the current time is within the pen non-use period supplies a waveform pattern for micro-vibration to the haptics driver **29** (step **S14**). It is sufficient if the waveform pattern supplied here is the same as the waveform pattern supplied in step **S4** of FIG. **4**. As a result, an electric signal for vibrating the vibrator **21b** with an amplitude so small that the user cannot recognize the vibration is supplied from the haptics driver **29** to the vibration device **21** (to both ends of the coil **21e**), and consequently, the vibrator **21b** cannot move freely, so that the vibration of the vibration device **21** caused by the impact generated when the pen tip is brought into contact with the panel surface **3a** is suppressed.

[0066] Next, the short-range wireless communication unit **28** determines whether the pen non-use period has continued for a predetermined time period or more (step **S15**). The short-range wireless communication unit **28** that has obtained a positive determination result in this determination returns to step **S10** and enters the sleep mode. On the other hand, the short-range wireless communication unit **28** that has obtained a negative determination result returns to step **S13** and continues the processing.

[0067] As described above, according to the position detecting system **1** of the present embodiment, when the startup mode has been entered and the current time is within the pen non-use period, an electrical signal corresponding to a waveform pattern for micro-vibration is supplied

to the vibration device **21**, so that the movement of the vibrator **21b** during the pen non-use period is suppressed. Hence, the vibration of the vibration device **21** caused by the impact generated when the pen tip is brought into contact with the panel surface **3a** can be suppressed.

[0068] Note that, according to the position detecting system **1** of the present embodiment, when the user brings the pen tip into contact with the panel surface **3a** while the short-range wireless communication unit **28** is in the sleep mode, the impact causes vibration of the vibration device **21**. However, while the user is writing text or drawing a picture, even if the user removes the pen tip from the panel surface **3a**, the short-range wireless communication unit **28** maintains the state of having entered the startup mode, as long as the removal is temporary, so that the position detecting system **1** of the present embodiment can sufficiently suppress vibration of the vibration device **21** that could be caused by the impact.

[0069] Next, the position detecting system **1** according to a third embodiment of the present disclosure will be described. The position detecting system **1** according to the present embodiment differs from the position detecting system **1** according to the first embodiment in the configuration and operation of the haptics driver **29** and the content of the control of the vibration device **21** executed by the short-range wireless communication unit **28**, but is otherwise similar to the position detecting system **1** according to the first embodiment. The position detecting system **1** according to the present embodiment will be described below, by focusing on the differences from the position detecting system **1** according to the first embodiment.

[0070] FIG. **7** is a diagram illustrating the functional blocks of the pen **2** according to the present embodiment. However, this diagram schematically illustrates only the parts related to the vibration device **21** and the haptics driver **29**. As illustrated in FIG. **7**, the haptics driver **29** includes an oscillator circuit **29a** and a switch element **29b** (drive switch) connected in series to the coil **21e** in the vibration device **21**.

[0071] The oscillator circuit **29a** is a circuit that generates an electric signal that oscillates according to a waveform pattern supplied from the short-range wireless communication unit **28** illustrated in FIG. **3**. The oscillator circuit **29a** is configured to continuously generate an electric signal that oscillates according to the waveform pattern for micro-vibration described in the first embodiment, and, configured to, when a waveform pattern is supplied from the short-range wireless communication unit **28**, switch the electric signal to be generated to an electric signal that oscillates according to the waveform pattern for a time period determined by the length of the waveform pattern. The switch element **29b** is a single-pole single-throw switch that is inserted between the vibration device **21** (coil **21e**) and the oscillator circuit **29a** and serves to switch the connection state between the two. The electric signal generated by the oscillator circuit **29a** is supplied to the coil **21e** when the switch element **29b** is on, whereas the electric signal is not supplied to the coil **21e** when the switch element **29b** is off.

[0072] FIG. **8** is a process flow diagram illustrating the control of the vibration device **21** executed by the short-range wireless communication unit **28** according to the present embodiment. As illustrated in FIG. **8**, the short-range wireless communication unit **28** first determines whether or not the current time is within the pen non-use period (step **S20**). The details of this determination are similar to those of step **S1** in FIG. **4**.

[0073] The short-range wireless communication unit **28** that has determined in step **S20** that the current time is not within the pen non-use period executes steps **S103** to **S106** and step **S5** as in the example of FIG. **4**, but the short-range wireless communication unit **28** in the present embodiment executes a process of control to turn on the switch element **29b** before executing step **S105** (step **S25**). This is a measure to allow an electric signal (an electric signal generated by the oscillator circuit **29a**) to flow through the coil **21e**. When the switch element **29b** is on at a stage prior to executing step **S25**, the short-range wireless communication unit **28** may skip step **S25**. This also applies to steps **S23**, **S24**, and **S34** to be described later.

[0074] On the other hand, the short-range wireless communication unit **28** that has determined in

step S20 that the current time is within the pen non-use period performs finger detection by using the capacitive sensor 27 (step S21), and determines whether a finger has been detected as a result (step S22). The short-range wireless communication unit 28 that has determined that a finger has not been detected turns off the switch element 29b (step S24), and then returns to step S20 to continue the processing.

[0075] On the other hand, the short-range wireless communication unit 28 that has determined in step S22 that a finger has been detected, turns on the switch element 29b (step S23), and then returns to step S20 to continue the processing. As a result, since an electric signal (an electric signal that is generated by the oscillator circuit 29a and oscillates according to a waveform pattern for micro-vibration) flows through the coil 21e, the vibration of the vibration device 21 is suppressed, as in the first embodiment.

[0076] As described above, according to the position detecting system 1 of the present embodiment, when the current time is within the pen non-use period and a finger is being detected by the capacitive sensor 27, an electric signal (an electric signal that is generated by the oscillator circuit 29a and oscillates according to a waveform pattern for micro-vibration) flows through the coil 21e, so that the movement of the vibrator 21b is suppressed during the pen non-use period. Hence, the vibration of the vibration device 21 caused by the impact generated when the pen tip is brought into contact with the panel surface 3a can be suppressed.

[0077] In addition, in the position detecting system 1 according to the present embodiment, as in the position detecting system 1 according to the second embodiment, the capacitive sensor 27 does not have to be provided in the pen 2 and steps S21 and S22 in FIG. 8 may be omitted, while still obtaining the same effects as described above.

[0078] In the position detecting system 1 according to the present embodiment, the generation of the electric signal by the oscillator circuit 29a itself may be turned on and off instead of the on/off control of the switch element 29b. In this way, in addition to the above described effects, the effect of reducing the power consumption of the pen 2 can also be obtained.

[0079] Next, the position detecting system 1 according to a fourth embodiment of the present disclosure will be described. The position detecting system 1 according to the present embodiment differs from the position detecting system 1 according to the third embodiment in that the pen 2 does not have the capacitive sensor 27, the short-range wireless communication unit 28 has a sleep mode, and that the content of the control of the vibration device 21 executed by the short-range wireless communication unit 28 is different. Otherwise, the present embodiment is similar to the position detecting system 1 according to the third embodiment. Below, the position detecting system 1 according to the present embodiment will be described, by focusing on the differences from the position detecting system 1 according to the third embodiment.

[0080] The functional blocks of the pen 2 and the electronic device 3 according to the present embodiment are similar to those according to the second embodiment illustrated in FIG. 5, except that the haptics driver 29 has the configuration illustrated in FIG. 7. The short-range wireless communication unit 28 according to the present embodiment is configured to operate in either the startup mode or the sleep mode, similarly to the short-range wireless communication unit 28 according to the second embodiment. Details of each mode are as described in the second embodiment. However, in the sleep mode according to the present embodiment, the switch element 29b in the haptics driver 29 is turned off.

[0081] FIG. 9 is a process flow diagram illustrating the control of the vibration device 21 executed by the short-range wireless communication unit 28 according to the present embodiment. As illustrated in FIG. 9, the short-range wireless communication unit 28 according to the present embodiment first enters a sleep mode (step S30), and repeatedly executes a determination as to whether or not a predetermined startup operation has been executed, until a positive determination result is obtained (step S31). At this time, the switch element 29b is off.

[0082] In the case where a positive determination result is obtained in step S31, the short-range

wireless communication unit **28** enters a startup mode (step **S32**) and determines whether or not the current time is within the pen non-use period (step **S33**). The details of this determination are similar to those in step **S1** of FIG. **4**.

[0083] The short-range wireless communication unit **28** that has determined in step **S33** that the current time is not within the pen non-use period executes the processes of steps **S103** to **S106**, step **S5**, and step **S25** as in the example of FIG. **8**. The details of this point are as described with reference to FIGS. **4** and **8**.

[0084] On the other hand, the short-range wireless communication unit **28** that has determined in step **S33** that the current time is within the pen non-use period performs control to turn on the switch element **29b** (step **S34**). As a result, an electric signal (an electric signal that is generated by the oscillator circuit **29a** and oscillates according to a waveform pattern for micro-vibration) flows through the coil **21e**, so that the vibration of the vibration device **21** is suppressed as described above. Thereafter, the short-range wireless communication unit **28** determines whether the pen non-use period has continued for a predetermined time period or more (step **S35**). When a positive determination result is obtained, the short-range wireless communication unit **28** returns to step **S30** and enters the sleep mode. At this time, the switch element **29b** returns to off. On the other hand, the short-range wireless communication unit **28** that has obtained a negative determination result returns to step **S33** and continues the processing.

[0085] As described above, according to the position detecting system **1** of the present embodiment, when the startup mode has been entered and the current time is within the pen non-use period, an electric signal (an electric signal that is generated by the oscillator circuit **29a** and oscillates according to a waveform pattern for micro-vibration) flows through the coil **21e**, and the movement of the vibrator **21b** is suppressed during the pen non-use period. Therefore, the vibration of the vibration device **21** caused by the impact generated when the pen tip is brought into contact with the panel surface **3a** can be suppressed.

[0086] It should be noted that, according to the position detecting system **1** of the present embodiment, when the user brings the pen tip into contact with the panel surface **3a** while the short-range wireless communication unit **28** is in the sleep mode, the impact causes vibration of the vibration device **21** as in the position detecting system **1** of the second embodiment. However, even if the user removes (separates) the pen tip from the panel surface **3a** while writing text or drawing a picture, since the short-range wireless communication unit **28** maintains the state of having entered the startup mode as long as the detachment is temporary, the position detecting system **1** of the present embodiment can sufficiently suppress vibration of the vibration device **21** that could be caused by the impact.

[0087] Although the embodiments of the present disclosure have been described above, the present disclosure is not limited to these embodiments, and can be embodied in various further embodiments based on the present disclosure.

[0088] For example, in each of the above embodiments, an example is described in which the short-range wireless communication BT is used to transmit identification information that identifies a waveform pattern from the electronic device **3** to the pen **2**. However, the identification information may be transmitted using uplink signal **US**, for example.

Claims

1. A pen, comprising: a vibration device including a vibrator; and a processing unit that suppresses movement of the vibrator during a pen non-use period during which no pen input is being performed.
2. The pen according to claim 1, wherein the processing unit is configured to: supply a first electrical signal to the vibration device to vibrate the vibration device while a user is writing with the pen, and suppress the movement of the vibrator during the pen non-use period, by supplying

during the pen non-use period a second electrical signal to the vibration device, the second electrical signal oscillating with an amplitude smaller than that of the first electrical signal.

3. The pen according to claim 2, wherein the vibration device includes a coil, and the processing unit supplies the second electrical signal to the vibration device by supplying the second electrical signal to both ends of the coil.
 4. The pen according to claim 1, comprising: an oscillator circuit; and a switch element inserted between the vibration device and the oscillator circuit, wherein the vibration device includes a coil, and the processing unit turns on the switch element during the pen non-use period to allow an electrical signal to flow through the coil to thereby suppress the movement of the vibrator during the pen non-use period.
 5. The pen according to claim 4, wherein the electrical signal is generated by the oscillator circuit.
 6. The pen according to claim 1, comprising: a capacitive sensor arranged on a surface of the pen, wherein the processing unit suppresses the movement of the vibrator when a current time is within the pen non-use period and a finger is being detected by the capacitive sensor.
 7. The pen according to claim 1, wherein the processing unit is configured to: enter a startup mode in response to a predetermined startup operation, and to enter a sleep mode when a state in which no pen input is performed continues for a predetermined time period or more after entering the startup mode, and suppress the movement of the vibrator when the processing unit has entered the startup mode and a current time is within the pen non-use period.
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